CampusConnect

One Platform for Students, Faculty & Campus Life

Product Requirements Document
Version: 1.0
Status: Approved for Development
Target Audience: Engineering, Design, Stakeholders

Table of Contents

(Note: To render the Table of Contents, click inside this area, press Ctrl+A to select all, and press F9. Alternatively, right click and select 'Update Field'.)

1. Executive Summary

1.1 Product Vision

CampusConnect aims to become the single digital platform that manages every aspect of university life. Modern higher education institutions suffer from massive digital fragmentation. Students and faculty are forced to context-switch between disparate systems for academics, attendance tracking, event discovery, club management, campus navigation, and critical announcements. This fragmented architecture leads to poor user experiences, data silos, and massive administrative overhead.

CampusConnect solves this by bringing everything together into one unified, secure application with deeply integrated, role-based experiences tailored specifically for Students, Faculty, and Administrators. By consolidating these discrete workflows into a centralized ecosystem, our long-term vision is to drastically improve operational efficiency, deeply enhance student engagement, eliminate redundant administrative workloads, and foster a truly connected digital campus ecosystem.

1.2 Strategic Overview

CampusConnect is a production-grade University Management Platform built utilizing the modern MERN Stack. By abstracting the underlying complexity of university operations, the platform delivers a cohesive UI/UX that drives adoption and engagement.

The system supports three primary roles:

• Student: Empowered to manage their academic journey, discover campus culture, and track their performance via a personalized dashboard.

• Faculty: Equipped with high-leverage tools to manage attendance instantly, communicate effectively, and organize departmental events with zero friction.

• Administrator: Provided with a god's-eye view of campus operations, featuring robust analytics, user management, and global communication tools.

Ultimately, CampusConnect is not just an application; it is a digital transformation engine for higher education. It replaces fragmented IT infrastructure with a streamlined, scalable SaaS architecture.

2. Problem Statement

The current operational framework of most universities is plagued by deeply entrenched inefficiencies, severely impacting the daily lives of all stakeholders. The core problems we are addressing include:

2.1 Multiple Disconnected Portals

Problem: A typical student uses one portal for grades, a separate mobile app for attendance, Google Forms for event registrations, WhatsApp groups for club announcements, and a static PDF on a website for campus maps. This causes massive cognitive load, forgotten passwords, and missed communications.

Solution: CampusConnect aggregates these features. A student logs in once (SSO) and possesses immediate access to their entire campus life. The business value here is a massive increase in DAU and a decrease in IT support tickets related to access and credential recovery.

2.2 Manual & Inaccurate Attendance Management

Problem: Faculty lose 10-15 minutes per lecture conducting roll calls or passing around sign-in sheets. This process is highly susceptible to proxy attendance and manual data entry errors, delaying administrative intervention for at-risk students.

Solution: Introducing dynamic QR-based attendance. Faculty generate a time-expiring QR code; students scan it via the CampusConnect app. Attendance is logged securely and instantly, giving faculty their time back and providing real-time data to administrators.

2.3 Event & Club Management Inefficiencies

Problem: Campus events suffer from low turnout due to poor discovery mechanisms. Organizers rely on physical posters or disjointed social media posts. Tracking registrations via spreadsheets leads to chaotic check-ins.

Solution: A centralized Event and Club hub where students can discover, register, and receive digital tickets. Organizers get a dashboard to track RSVPs and scan tickets at the door, streamlining the end-to-end event lifecycle.

2.4 Communication Silos & Information Asymmetry

Problem: Critical announcements get buried in cluttered student email inboxes. Emergency notifications lack a centralized, high-visibility broadcast mechanism.

Solution: A prioritized Notification and Announcement engine. Administrators can push global, department-specific, or course-specific alerts that trigger immediate push notifications, ensuring a 90%+ read rate for critical updates.

3. Product Goals & Success Metrics

3.1 Primary Product Goals

- Create one unified university platform that replaces minimum 3 legacy tools.
- Digitize and automate manual campus operations (attendance, ticketing).

• Improve student engagement with campus life (events, clubs).
• Provide actionable, real-time analytics to university administration.
• Simplify campus navigation for new students and visitors.

3.2 Business Goals

• Reduce manual administrative hours by 40% within the first academic year.
• Increase total digital adoption across the student body to >95% within 3 months.
• Establish a scalable, multi-tenant architecture that enables future SaaS deployment to other universities.

3.3 Key Performance Indicators (KPIs)

Metric	Definition	Target (6 Months Post-Launch)
Daily Active Users (DAU)	Percentage of enrolled students logging in daily.	> 70%
Monthly Active Users (MAU)	Percentage of enrolled students logging in monthly.	> 95%
Attendance Submission Speed	Average time taken to log attendance for a class of 50.	< 30 seconds
Event Registrations	Total event RSVPs processed through the platform.	+50% vs Previous Year
Average Login Frequency	Number of sessions per user per week.	8 - 12 sessions
Notification Open Rate	Percentage of users viewing push/in-app announcements.	> 85%
System Uptime	Platform availability excluding scheduled maintenance.	99.9%
API Response Time	Average latency for core dashboard endpoints.	< 200ms

4. Target Users & Personas

Persona: Alex - Undergraduate Student

Age: 19
Goals: Maintain high attendance, stay updated on deadlines, join technical clubs.
Pain Points: Misses important club announcements on email. Forgets physical ID card. Has trouble finding new lecture halls.
Technical Skills: High (Digital Native)
Typical Daily Activities: Attends lectures, completes assignments, uses social media to find campus events.

How CampusConnect Improves Experience: Provides a single dashboard for their schedule, a digital ID on their phone, and a unified feed of club events they can register for in one tap.

Persona: Prof. Davis - Senior Faculty Member

Age: 45
Goals: Maximize teaching time, track student performance easily, manage department announcements.
Pain Points: Wastes 10 minutes every class calling roll. Dislikes navigating clunky legacy ERP systems to update grades.
Technical Skills: Moderate
Typical Daily Activities: Lectures 3 times a day, grades papers, holds office hours.
How CampusConnect Improves Experience: QR Attendance gives her 10 minutes of teaching time back. The intuitive faculty dashboard allows her to push announcements to her specific classes instantly.

Persona: Sarah - University Administrator

Age: 38
Goals: Ensure smooth campus operations, manage student onboarding, resolve logistical issues.
Pain Points: Has no clear view of overall campus engagement. Handles lost & found manually via a chaotic spreadsheet.
Technical Skills: Moderate to High
Typical Daily Activities: Manages user databases, issues communications, coordinates across departments.
How CampusConnect Improves Experience: Provides global analytics dashboards. Replaces the manual lost & found with a centralized, searchable digital ledger. Streamlines user management.

Persona: Dr. Patel - Department Head

Age: 52
Goals: Monitor faculty performance, ensure curriculum delivery, manage department-wide events.
Pain Points: Data is siloed; cannot easily see which classes have poor attendance records without requesting manual reports.
Technical Skills: Moderate
Typical Daily Activities: Reviews department metrics, approves events, manages faculty assignments.
How CampusConnect Improves Experience: Provides automated, real-time reports on department attendance and engagement metrics, allowing for proactive interventions.

Persona: John - Student Club Coordinator

Age: 21
Goals: Increase event turnout, manage club memberships efficiently.
Pain Points: Relies on Google Forms and cash payments. Cannot easily scan tickets at the door, causing bottlenecks.
Technical Skills: High
Typical Daily Activities: Plans events, markets them on Instagram, manages club spreadsheets.
How CampusConnect Improves Experience: Gets a dedicated Club Management portal. Can publish events, track RSVPs dynamically, and use the platform's scanner to check students in at the door.

Persona: Priya - First-Year Student

Age: 18
Goals: Navigate the massive campus, make friends, understand university procedures.
Pain Points: Constantly lost. Doesn't know who to ask for basic FAQs (library hours, WiFi setup).
Technical Skills: High
Typical Daily Activities: Navigating campus, attending orientation, asking seniors for help.
How CampusConnect Improves Experience: The Campus Map module provides exact locations. The FAQ Chatbot provides instant answers to procedural questions without needing to visit the admin office.

5. User Journeys

5.1 The Student Journey

```
[ Landing Page ] -> [ Google OAuth Login ] -> [ Personalized Dashboard ]
    |
    +--> [ View Attendance ] (Checks percentage against university minimums)
    |
    +--> [ Events Module ] -> Discovers 'Tech Hackathon' -> [ 1-Click Register ] -> [ Digital Ticket Generated ]
    |
    +--> [ Campus Map ] -> Searches for "Block C Auditorium" -> [ View Directions ]
    |
    +--> [ Notification Center ] -> Views alert: "Library closing early today"
    |
    +--> [ Profile ] -> [ Access Digital ID Badge ]
```

Explanation: The student journey is optimized for rapid consumption of information. The most critical data (Attendance, Upcoming Schedule) is front-loaded on the dashboard. Discovery loops (Events, Clubs) are frictionless to drive engagement.

5.2 The Faculty Journey

[Login] -> [Faculty Dashboard]
    |
    +--> [Select Active Class] -> [Generate QR Code] -> Project on screen -> Watch real-time scan count
    |
    +--> [Announcements] -> Selects 'Data Structures CS101' -> Types message -> [Publish & Send Push]
    |
    +--> [Feedback Module] -> Reviews anonymous mid-semester feedback from students

Explanation: The faculty journey optimizes for time-saving workflows. Generating attendance and broadcasting messages require fewer than 3 clicks. This aggressive friction reduction drives faculty adoption.

5.3 The Administrator Journey

[Login] -> [Global Admin Dashboard]
    |
    +--> [Analytics Hub] -> Views campus-wide DAU, Event Turnout, and Attendance averages
    |
    +--> [User Management] -> Bulk imports new semester students via CSV
    |
    +--> [Global Announcements] -> Pushes emergency "Weather Alert" to all active users
    |
    +--> [Lost & Found] -> Approves new item listing found by security desk

6. Core Features & Scope

We map platform capabilities against business value to define a strict, focused MVP.

Feature	Description	Business Value	Priority	MVP	Future
Authentication	SSO via Google OAuth &	Security, fast onboarding	P0	Yes	No

	JWT.				
Role Management	RBAC for Student, Faculty, Admin.	Data isolation & security	P0	Yes	No
QR Attendance	Dynamic QR generation and scanning.	Time savings, accurate records	P0	Yes	No
Dashboard	Role-specific widget landing page.	User retention, usability	P0	Yes	No
Events & Ticketing	Centralized discovery and RSVP.	Engagement, data collection	P1	Yes	No
Announcements	Targeted push and in-app notifications.	Communication efficiency	P1	Yes	No
Campus Map	Interactive mapping of campus POIs.	Usability, freshman experience	P2	Yes	No
Analytics	Admin data visualization (Chart.js).	Data-driven administration	P1	Yes	No
Digital ID	Phone-based student verification.	Cost reduction on physical cards	P2	Yes	No
Lost & Found	Centralized catalog of lost items.	Operational efficiency	P2	Yes	No
FAQ Chatbot	Automated procedural answers.	Support ticket deflection	P2	Yes	No
AI Attendance Prediction	ML model predicting dropouts.	Proactive retention	P3	No	Yes
Video Meetings	Native WebRTC integration.	Remote learning	P3	No	Yes

6.1 RICE Prioritization Framework

Decisions for the MVP scope were driven by the RICE framework (Reach x Impact x Confidence / Effort).

Feature	Reach	Impact (1-5)	Confidence	Effort	Score

				(Months)	
QR Attendance	100%	5	90%	1.5	300
Announcements	100%	4	95%	1	380
Events Module	80%	4	85%	2	136
AI Prediction	20%	4	60%	4	12
Campus Map	40%	3	90%	1	108

Analysis: Foundational features (Attendance, Announcements) score exceptionally high due to 100% reach and high impact on daily operations. Complex AI features require high effort and have low confidence in early stages, explicitly validating their removal from the MVP scope.

7. MVP Scope Definitions

7.1 What is in Version 1.0 (MVP) and Why?

The MVP focuses strictly on "Operational Hygiene" and "Student Engagement."
• Authentication & RBAC: Fundamental requirement. Without secure boundaries, the system is unviable.
• QR Attendance: The primary hook for faculty adoption. By solving their biggest daily pain point immediately, we guarantee platform usage.
• Announcements & Notifications: The primary hook for administrators.
• Events, Clubs, Map, Chatbot: The primary hooks for student engagement.
We must deploy a holistic ecosystem in V1. If we only deploy attendance, it is just another admin tool. By including Events and Maps, it becomes a student-centric platform.

7.2 What is deferred and Why?

• Timetable Generator & Course Registration: These are deeply complex workflows tied to legacy ERP systems and strict compliance logic. Integrating these in V1 introduces massive risk and delays Time-to-Market.
• AI Academic Advisor / Prediction: Requires historical data to train models. V1 must act as the data collection engine before AI features can provide actual value.
• Fee Management: Handling financial transactions requires extensive security audits, payment gateway integration (Razorpay/Stripe), and liability handling. Postponed to V2 to ensure initial launch velocity.

8. Product Architecture & Workflows

The platform leverages a streamlined event-driven architecture to ensure minimal latency.

```
[Client Application - React/Vite]
     |
```

```
 1. User Action (e.g., Scan QR)
     |
     v
[API Gateway / Auth Middleware - Express.js]
     |
 2. Validate JWT & Role Permissions
     |
     v
[Business Logic Layer - Node.js Services]
     |
 3. Execute Transaction (e.g., Log Attendance)
     |
     v
[Data Layer - MongoDB Atlas] <---> [Redis Cache (Optional for V2)]
     |
 4. Write Data / Trigger Async Event
     |
     v
[Notification Service] ---> (Push to Client)
     |
 5. Return 200 OK Response to Client UI
```

9. Functional Requirements

This section translates product strategy into actionable engineering requirements.

Module: Authentication & Authorization

Purpose: Securely authenticate users and route them to their role-specific experiences.
Business Value: Prevents unauthorized access and protects sensitive FERPA/academic data.
Primary User Story: As a user, I want to log in using my university Google account so I don't have to remember another password.
Acceptance Criteria:

- System must support Google OAuth 2.0.
- JWT must expire after 24 hours.
- Invalid roles must be redirected to an error boundary.

Module: QR Attendance System

Purpose: Allow faculty to rapidly take attendance using dynamic visual codes.
Business Value: Recovers ~15 minutes of instructional time per class. Generates perfect

digital audit trails.
Primary User Story: As a faculty member, I want to display a QR code on the projector so that 100 students can mark themselves present simultaneously.
Acceptance Criteria:
- QR code must rotate/refresh every 10 seconds to prevent students sending photos to absent friends.
- System must geofence check-ins (optional MVP+) to campus IPs.
- Student app must render success state within 1 second of scan.

Module: Events & Ticketing

Purpose: Provide a centralized hub for discovering and registering for campus activities.
Business Value: Increases student engagement scores. Provides organizers with accurate headcount data for budget allocation.
Primary User Story: As a student, I want to browse upcoming events and tap 'Register' to secure my spot.
Acceptance Criteria:
- Users can filter events by category (Tech, Cultural, Sports).
- Registration generates a unique QR ticket linked to the student ID.
- Organizers can export a CSV of attendees.

Module: Global Announcements

Purpose: Replace email blasts with high-visibility digital notifications.
Business Value: Ensures critical information reaches the student body instantly.
Primary User Story: As an administrator, I want to push a priority alert so that students immediately know about a campus closure.
Acceptance Criteria:
- Admins can target announcements by Department, Year, or Global.
- High-priority announcements trigger mobile push notifications.
- System tracks 'Read' status for analytics.

Module: Campus Map

Purpose: Digitize the physical campus layout.
Business Value: Drastically improves the freshman and visitor experience, reducing frustration.
Primary User Story: As a new student, I want to search for 'Block B Library' and see it on a map.
Acceptance Criteria:
- Map must utilize Leaflet.js rendering.
- Includes search bar with fuzzy matching for building names.

- Pins show metadata (Opening hours, Department).

Module: Lost & Found

Purpose: Digitize the legacy cardboard box at the security desk.
Business Value: Improves recovery rate of expensive items (laptops, phones), building trust in campus administration.
Primary User Story: As a student, I want to report my lost laptop so that anyone who finds it can easily notify me.
Acceptance Criteria:

- Users can upload photos of found items.
- Admins must approve listings before they go public to prevent spam.
- Claiming an item initiates a secure direct message to the finder.

Module: FAQ Chatbot

Purpose: Automate responses to repetitive logistical queries.
Business Value: Deflects hundreds of tier-1 support emails away from administrative staff.
Primary User Story: As a user, I want to ask 'What are the library hours?' and get an instant answer.
Acceptance Criteria:

- Chatbot uses a predefined decision tree/NLP matching for MVP.
- Fallback to 'Contact Admin' if query is unresolved.
- Maintains a history of the current session.

Module: Analytics Dashboard

Purpose: Provide a real-time pulse of the university's operations.
Business Value: Enables data-driven decision making for university leadership.
Primary User Story: As a Dean, I want to see a chart of overall attendance trends to identify struggling departments.
Acceptance Criteria:

- Dashboard must render using Chart.js.
- Displays DAU, overall attendance %, and active events.
- Data must load in under 2 seconds.

10. Comprehensive User Stories

The following Epics and User Stories define the exact capabilities required for the MVP.

US-001: As a student, I want to view my current semester schedule on my dashboard so I know where my next class is.

US-002: As a student, I want to scan a QR code to mark my attendance so I don't have to wait for a roll call.

US-003: As a student, I want to view my overall attendance percentage so I can ensure I meet the 75% university requirement.

US-004: As a student, I want to browse a catalog of active campus clubs so I can find extracurriculars that match my interests.

US-005: As a student, I want to RSVP to a tech talk so my spot is reserved.

US-006: As a student, I want a digital copy of my ID card in the app so I can enter the library if I forget my physical wallet.

US-007: As a student, I want to receive push notifications for cancelled classes so I don't commute unnecessarily.

US-008: As a student, I want to browse the Lost & Found ledger so I can see if anyone found my keys.

US-009: As a student, I want to search the campus map for the 'Finance Office' so I can pay my dues easily.

US-010: As a student, I want to submit anonymous feedback regarding a course so the administration can improve teaching quality.

US-011: As a faculty member, I want to log in and instantly see the classes I am scheduled to teach today.

US-012: As a faculty member, I want to generate a dynamic QR code for attendance so I can process 100 students in 1 minute.

US-013: As a faculty member, I want to manually override attendance for a student who has a broken phone camera.

US-014: As a faculty member, I want to view an attendance report for my entire class to identify chronically absent students.

US-015: As a faculty member, I want to send an announcement strictly to my CS101 section regarding a changed assignment deadline.

US-016: As a faculty member, I want to create a department-level event for a guest lecture I am hosting.

US-017: As a faculty member, I want to review anonymous feedback submitted by my students.

US-018: As an administrator, I want a high-level dashboard showing total active users today to gauge platform adoption.

US-019: As an administrator, I want to bulk-upload student profiles via CSV to provision accounts for the new semester.

US-020: As an administrator, I want to create, edit, and delete user profiles to handle role changes or dropouts.

US-021: As an administrator, I want to manage the master list of Subjects and Departments to maintain data integrity.

US-022: As an administrator, I want to push a global emergency announcement that overrides user notification preferences.

US-023: As an administrator, I want to approve or reject Lost & Found submissions to ensure no inappropriate content is posted.

US-024: As an administrator, I want to view engagement metrics for campus events to see which clubs deserve more funding.

US-025: As an administrator, I want to review system-wide feedback to identify systemic issues with campus facilities.

11. Non-Functional Requirements (NFRs)

NFRs define system attributes such as performance, security, and usability. They are as critical as functional features.

11.1 Performance & Scalability

• Concurrency: The system must comfortably handle high burst traffic. Specifically, during the first 10 minutes of a lecture hour (e.g., 9:00 AM), thousands of students will scan attendance QR codes simultaneously. The API must resolve attendance mutations in < 500ms under a load of 5,000 concurrent requests.

• Database Optimization: MongoDB collections must be properly indexed, particularly on `userId`, `eventId`, and `attendanceDate` fields to prevent slow queries during dashboard load.

• Scalability: The backend (Node/Express) must be stateless to allow horizontal scaling via containerization (Docker/Kubernetes) during peak exam or registration weeks.

11.2 Security & Compliance

• Authentication: All API endpoints (except public login/maps) must be protected via JWT validation in the middleware.

• Authorization: Strict Role-Based Access Control (RBAC). A Student token attempting to hit an Administrator endpoint (e.g., `/api/admin/users`) must immediately return a 403 Forbidden.

• Data Protection: Passwords (if used alongside OAuth) must be hashed using bcrypt with a minimum salt round of 10. No plaintext PII (Personally Identifiable Information) should be unnecessarily exposed in GET requests.

• Rate Limiting: Implement IP-based and User-based rate limiting to prevent brute-force attacks on login endpoints and spam on the feedback/lost-and-found forms.

11.3 Usability & Accessibility

• Responsive Design: The UI must be mobile-first. 85%+ of student interaction will occur on mobile devices. The web application must adapt flawlessly to iOS and Android viewport sizes.

• Accessibility (a11y): The application should strive for WCAG 2.1 AA compliance. Forms must have proper ARIA labels, and color contrast (especially in Dark Mode) must pass accessibility standards to support visually impaired students.

12. Risks & Mitigation Strategies

Identified Risk	Impact Level	Mitigation Strategy
QR Attendance Fraud (Students sharing codes off-campus)	High	Implement 10-second rotating QR codes. Future phase: Implement Geofencing via device location API.
Low Faculty Adoption	High	Conduct comprehensive onboarding sessions. Highlight the immediate time-saving benefits of the dashboard.
Database Bottlenecks during peak hours	Medium	Implement Redis caching for high-read, low-write endpoints like Announcements and Map data.
Notification Fatigue	Medium	Implement strict categorization. Allow students to opt-out of club alerts, but enforce mandatory reception of academic/global alerts.
Inappropriate content in Lost & Found	Low	Implement an admin approval queue before any

		user-generated content becomes publicly visible.

13. Product Roadmap

Our strategic rollout ensures we capture baseline operational value in V1 before expanding into advanced administrative and AI-driven features.

Phase 1: Validation & Core Operations (MVP - Months 1-3)

Focus: Replacing manual processes and establishing the digital ecosystem.

- Complete Auth & RBAC
- QR Attendance Engine
- Centralized Events & Clubs
- Global Announcements & Notifications
- Campus Map & Basic Chatbot
- Admin Analytics Dashboard

Phase 2: Academic Deepening & Monetization Prep (Months 4-8)

Focus: Tying the platform directly to heavy university infrastructure.

- Fee Management & Payment Gateway (Razorpay/Stripe)
- Hostel / Dormitory Management Module
- Timetable Generation Interface
- Parent Portal (Read-only access to attendance/grades)

Phase 3: AI & Smart Campus Ecosystem (Months 9-18)

Focus: Moving from descriptive analytics to predictive analytics.

- AI Academic Advisor (LLM integration for course selection)
- AI Attendance Prediction (Flags high-risk students to counselors before they drop out)
- Smart Campus IoT (Integration with library turnstiles and smart parking)
- Native Mobile Apps (React Native iOS/Android builds for better push notification reliability)

14. High-Level Technical Overview

CampusConnect is architected on the MERN stack, chosen for its ubiquitous developer ecosystem, JSON-native data handling, and excellent performance characteristics for real-time, I/O heavy web applications.

• Frontend: React.js bundled with Vite for rapid HMR and optimized builds. UI styling leverages modern utility frameworks (Tailwind/Bootstrap) for responsive, consistent design language. Data visualization powered by Chart.js. Mapping powered by Leaflet.js.

• Backend: Node.js running Express.js. Acts as a robust RESTful API gateway. Handles authentication, business logic validation, and database interfacing.

• Database: MongoDB Atlas. A NoSQL document database perfectly suited for the dynamic, hierarchical nature of university data (e.g., embedding multiple attendance records within a specific course document).

• Infrastructure: Frontend deployed on Vercel/Netlify globally distributed CDNs. Backend hosted on Render/AWS. This fully managed cloud stack eliminates the need for on-premise university server maintenance.

15. Appendix

15.1 Glossary

• DAU: Daily Active Users.
• RBAC: Role-Based Access Control.
• MERN: MongoDB, Express, React, Node.js.
• JWT: JSON Web Token, used for stateless secure session management.
• SSO: Single Sign-On.

15.2 Assumptions

• The university possesses reliable campus-wide WiFi, allowing students to access the platform during lectures to mark attendance.
• Students own smartphones capable of rendering modern web applications and scanning QR codes.
• The university administration is willing to mandate the platform's use for official communication.

--- End of Product Requirements Document ---